



PES

UNIVERSITY

CELEBRATING 50 YEARS

PROGRAMMING WITH PYTHON

Lekha A

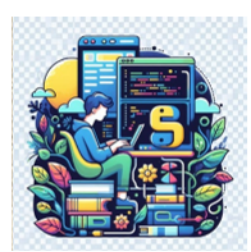
Computer Applications

PROGRAMMING WITH PYTHON

Python Basics and Data Structures Review of Constructs

Lekha A

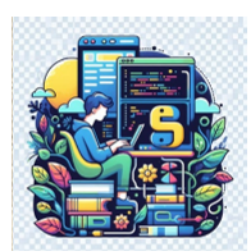
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PROGRAMMING WITH PYTHON

Recap

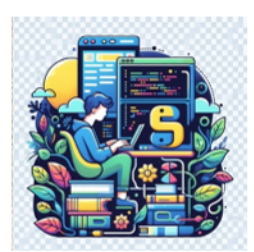
- Review of Data Types



PROGRAMMING WITH PYTHON

Conditions

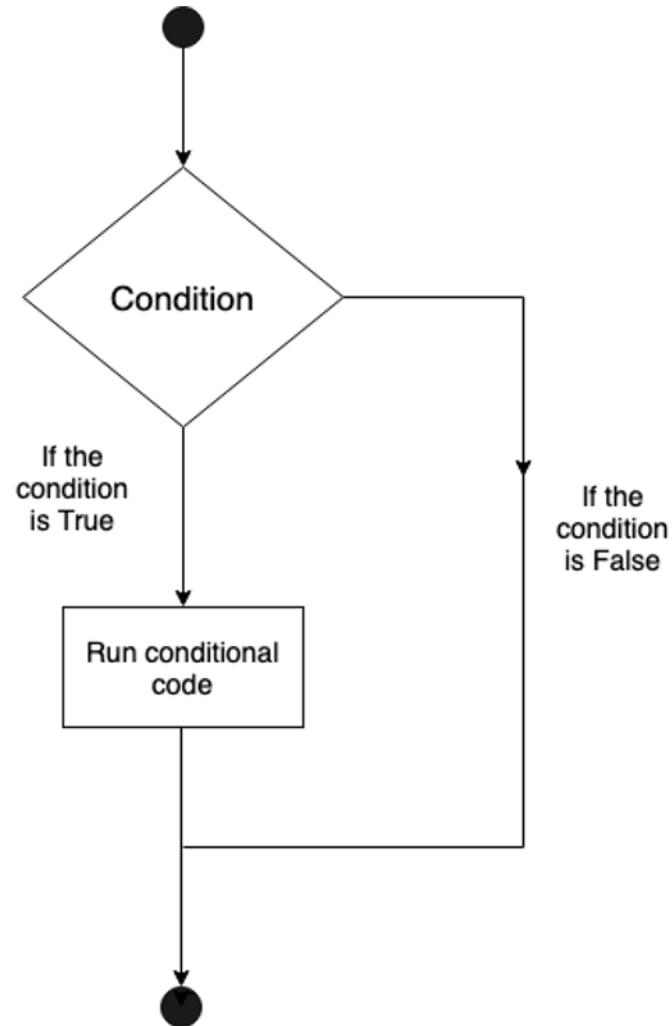
- Python Decision Making
 - if statements
 - if...else statements
 - elif Statement
 - nested if statements
 - Single Statement Suites

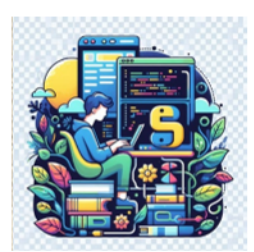


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If statement

- Syntax
 - if test expression:
 - statement(s)

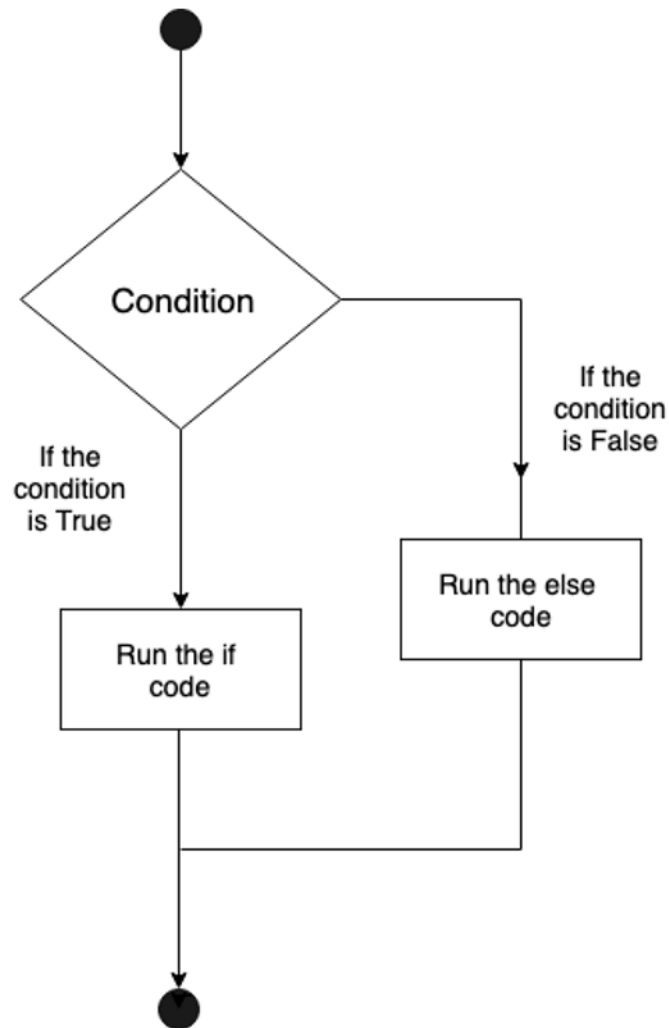


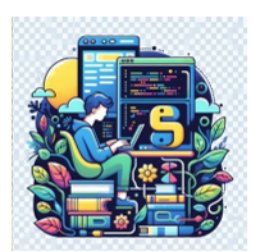


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If-else statement

- Syntax
 - if test expression:
 - Body of if
 - else:
 - Body of else

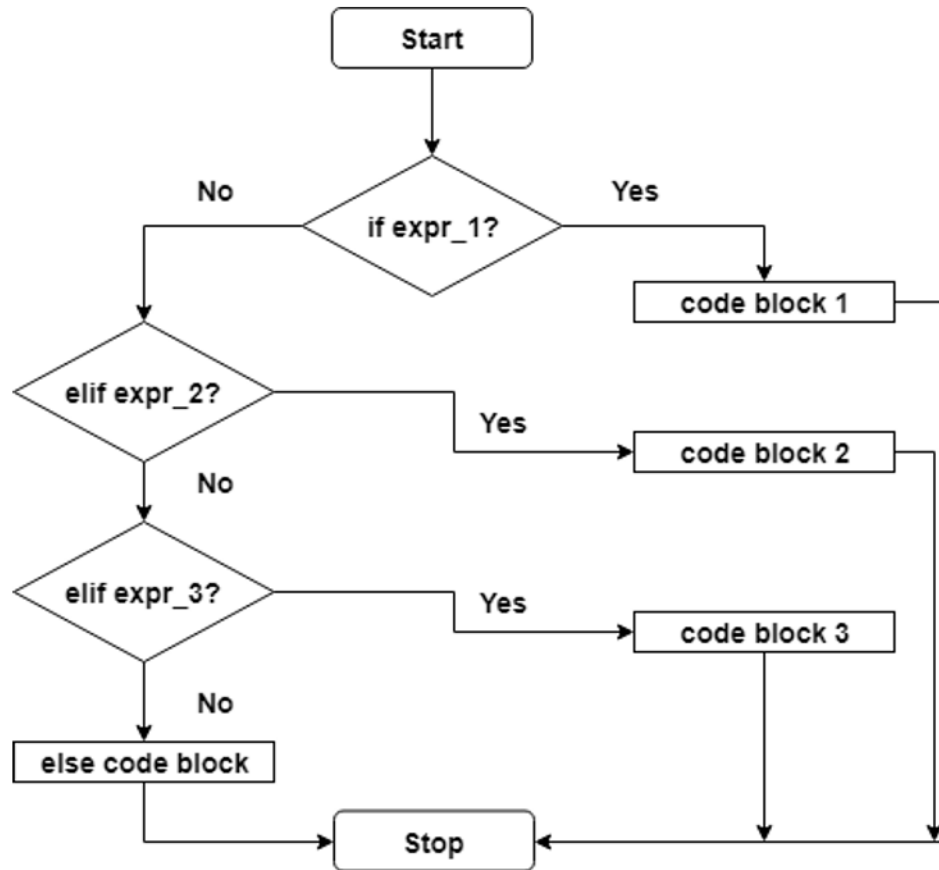


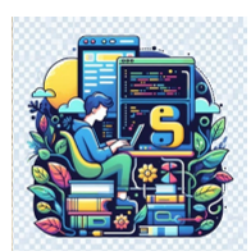


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Elif Statement

- Syntax
 - if expression1:
 - statement(s)
 - elif expression2:
 - statement(s)
 - elif expression3:
 - statement(s)
 - else:
 - statement(s)

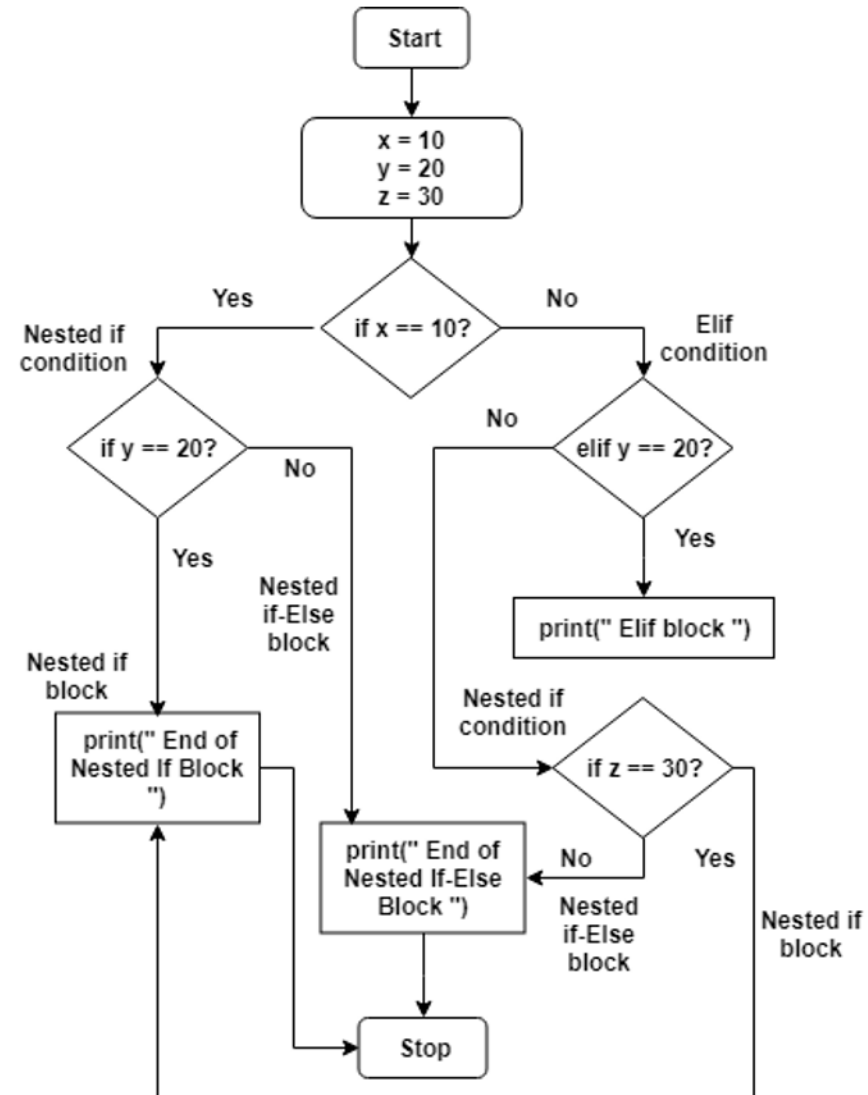


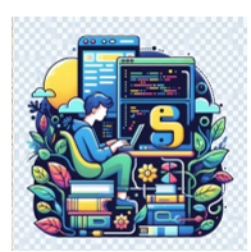


PROGRAMMING WITH PYTHON

Nested if Statement

```
if Logical_Expression_1 :  
    if Logical_Expression_1.1 :  
        if Logical_Expression_1.1.1 :  
            Indented Code Block 1.1.1  
        else  
            Indented Code Block  
    elif Logical_Expression_1.2 :  
        Indented Code Block 1.2  
    else :  
        Indented Code Block  
elif Logical_Expression_2 :  
    Indented Code Block 2  
elif Logical_Expression_3 :  
    Indented Code Block 3  
...  
else :  
    Indented Code Block
```





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Single Statement

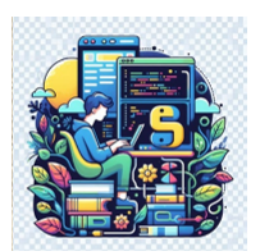
- If expression_1: statement



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Loops



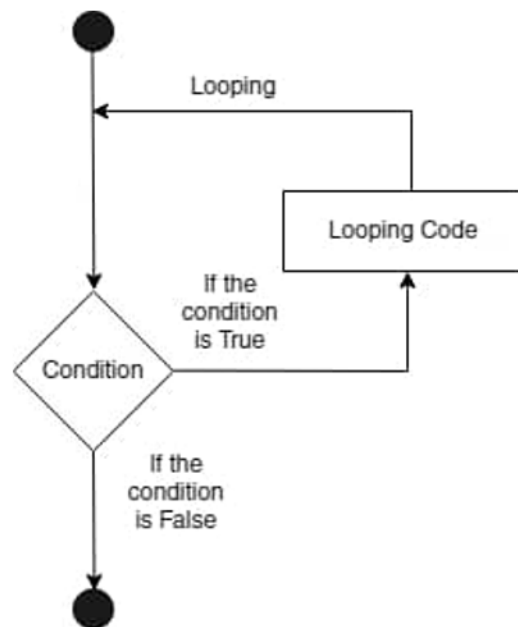


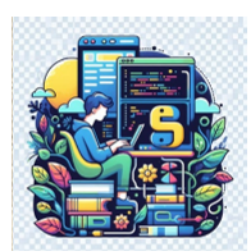
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While loop

- The while loop in Python is used to iterate over a block of code as long as the test expression (condition) is true.

- **while test-expression:**
- **statement**

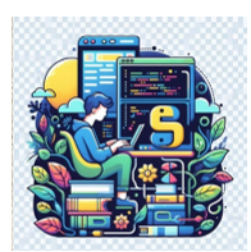




PROGRAMMING WITH PYTHON

While loop with else

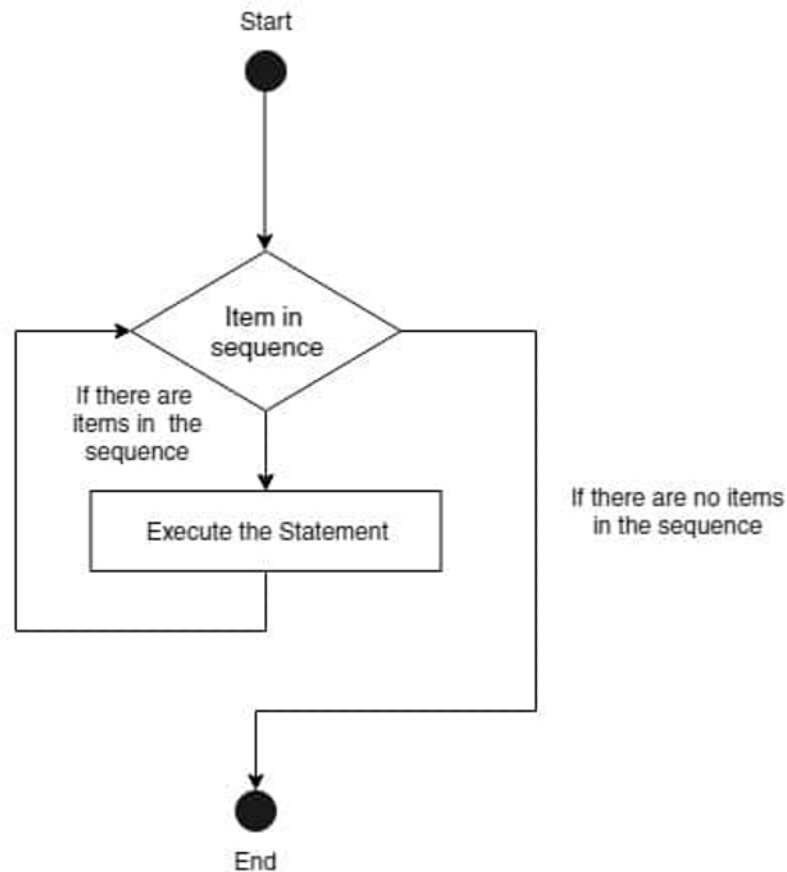
- 'while' can have an optional 'else' block.
- The else part is executed if the condition in the while loop evaluates to False.
 - while test_expression:
 - body
 - else:
 - statement

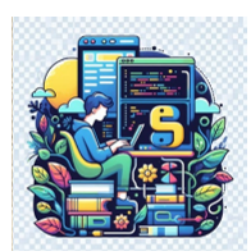


PROGRAMMING WITH PYTHON

For loop

- The for loop in Python is used to iterate over a sequence (list, tuple, string) or other iterable objects.
- Iterating over a sequence is called
- traversal.

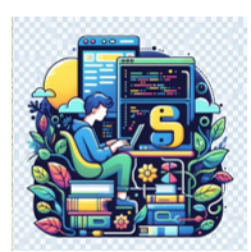




PROGRAMMING WITH PYTHON

For loop

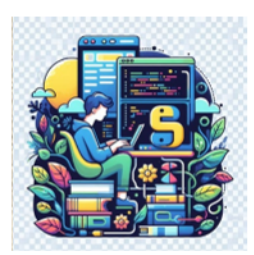
- Iterating over a sequence is called traversal.
 - for val in sequence:
 - Body of for
 - val is the variable that takes the value of the item inside the sequence on each iteration.



PROGRAMMING WITH PYTHON

For loop

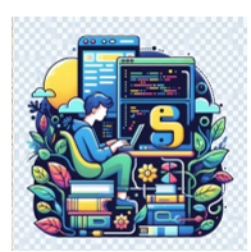
- `range(start, stop[, step])`
 - It takes three arguments.
 - Out of the three 2 arguments are optional. I.e., start and step are the optional arguments.



PROGRAMMING WITH PYTHON

For loop with range

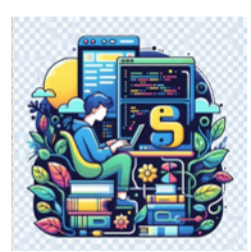
- A start argument is a starting number of the sequence. i.e., lower limit.
 - By default, it starts with 0 if not specified.
- A stop argument is an upper limit. i.e., generate numbers up to this number.
 - The range() doesn't include this number in the result.
- The step is a difference between each number in the result.
 - The default value of the step is 1 if not specified.



PROGRAMMING WITH PYTHON

For loop with range

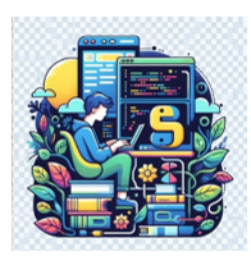
- `for var in range():`
- statements using var as index



PROGRAMMING WITH PYTHON

For loop with else

- A for loop can have an optional else block as well.
- The else part is executed if the items in the sequence used in for loop exhausts.
 - for val in sequence:
 - Body of for
 - else:
 - statement

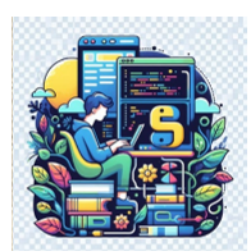


PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `if (1, 2):` Indentation error

 `print('foo')`
- `if (1, 2): print('Python')` Python



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `x = 10`

`if x > 5:`

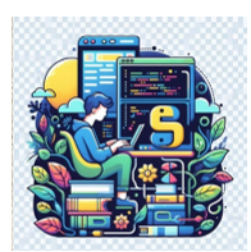
`print("Greater than 5")`

`elif x > 8:`

`print("Greater than 8")`

`else:`

`print("Less than or equal to 5")`



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `x = 10`

Greater than 5

`if x > 5:`

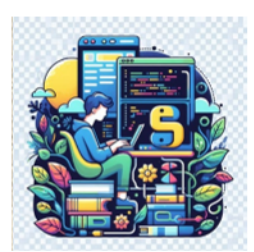
`print("Greater than 5")`

`elif x > 8:`

`print("Greater than 8")`

`else:`

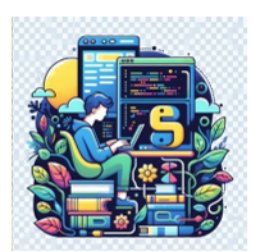
`print("Less than or equal to 5")`



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `for i in range(5):`
 - `0`
 - `1`
 - `if i == 3:`
 - `2`
 - `4`
 - `continue`
 - `print(i)`



PROGRAMMING WITH PYTHON

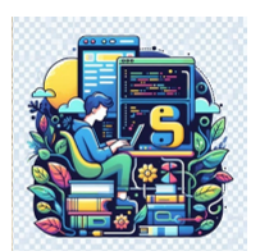
Review of Conditions and Loops

- if 'bar' in {'foo': 1, 'bar': 2, 'baz': 3}:
1
2
print(1)
4
print(2)

if 'a' in 'qux':

print(3)

print(4)



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `d = {'a': 0, 'b': 1, 'c': 0}` ok

`if d['a'] > 0: print('ok')`

`elif d['b'] > 0: print('ok')`

`elif d['c'] > 0: print('ok')`

`elif d['d'] > 0: print('ok')`

`else: print('not ok')`



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `d = {'a': 0, 'b': 1, 'c': 0}` ok

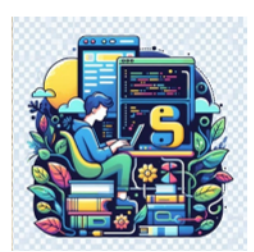
`if d['a'] > 0: print('ok')`

`elif d['b'] > 0: print('ok')`

`elif d['c'] > 0: print('ok')`

`elif d['d'] > 0: print('ok')`

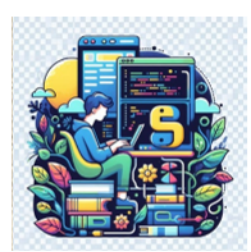
`else: print('not ok')`



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `for num in range(-2,-5,-1):` `-2, -3, -4,`
`print(num, end=", ")`
- `for l in 'Jhon':` `J, h, o, n,`
`if l == 'o':`
`pass`
`print(l, end=", ")`



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `x = 0; a = 0; b = -5`

2

`if a > 0:`

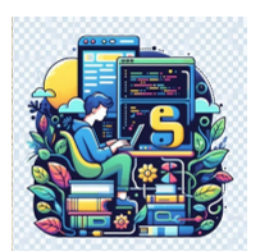
`if b < 0:        x = x + 5`

`elif a > 5:      x = x + 4`

`else:          x = x + 3`

`else:  x = x + 2`

`print(x)`

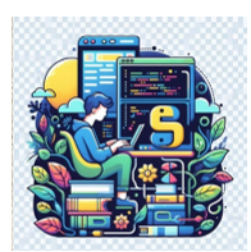


PROGRAMMING WITH PYTHON

Review of Conditions and Loops

21

- `var = 10`
- `for i in range(10):`
- `for j in range(2, 10, 1):`
- `if var % 2 == 0:`
- `continue; var += 1`
- `var+=1`
- `else: var+=1`
- `print(var)`



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- for num in range(10, 14):
10
11
for i in range(2, num):
12
if num%i == 1:
13

print(num)

break



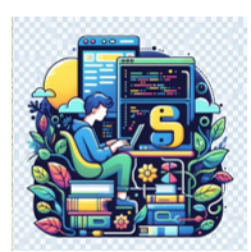
- `for num in range(2,-5,-1):` `2, 1, 0, -1, -2, -3, -4,`

 `print(num, end=", ")`
- `x = 0` `1 ..... 90`

 `for i in range(10):`

 `for j in range(-1, -10, -1):`

 `x += 1; print(x)`



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `a, b = 12, 5`

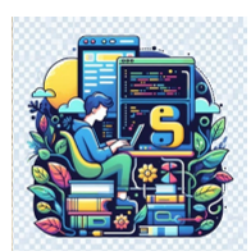
True

```
if a + b:
```

```
    print('True')
```

```
else:
```

```
    print('False')
```



PROGRAMMING WITH PYTHON

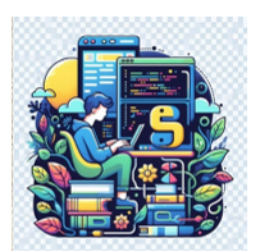
Review of Conditions and Loops

- `x = 0` 100

`while (x < 100):`

`x+=2`

`print(x)`



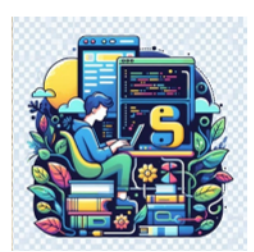
PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `numbers = [10, 20]`
`items = ["Chair", "Table"]`
`for x in numbers:`

`for y in items:`

`print(x, y)`
- | |
|----------|
| 10 Chair |
| 10 Table |
| 20 Chair |
| 20 Table |



PROGRAMMING WITH PYTHON

Review of Conditions and Loops

- `x = 0; a = 5; b = 5`

3

if `a > 0`:

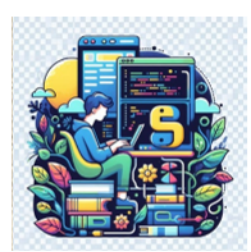
 if `b < 0`: `x = x + 5`

 elif `a > 5`: `x = x + 4`

 else: `x = x + 3`

else: `x = x + 2`

`print(x)`



PROGRAMMING WITH PYTHON

Recap

- Conditional Constructs
- Loops
- Review



THANK YOU

Lekha A
Computer Applications